

1) Determinazione specifiche di progetto

Incognite: G_F ; K_c da ε_r ; K_c da d_a e/o d_p ; ω_c da d_p e/o d_s ; ζ , $T_{p,0}$, $S_{p,0}$ da $\hat{s}$; ω_c da t_r ; ω_c da $t_{s,\varepsilon\%}$; ν da d_a e/o d_p ; M_S^{LF} , M_T^{HF} .

a) Calcolo G_F

$$G_F = \frac{1}{G_s K_d} \quad \text{se } \nu + p > 0$$

b) Calcolo ν e K_c da ε_r

se $\varepsilon_r = 0 \rightarrow \nu > h - p$, $|K_c| = \forall$

se $\varepsilon_r \neq 0 \rightarrow \nu \geq h - p$

$$|K_c| \geq \begin{cases} \left| \frac{R_0 K_d^2}{\varepsilon_r K_p G_a} \right| & \text{se } \nu + p \neq 0 \\ \frac{R_0 K_d^2}{\varepsilon_r} - K_d & \text{se } \nu + p = 0 \end{cases}$$

K_p = guadagno di G_p in cost. tempo. (p : poli nell'origine di G_p espresso in cost. tempo)

c) Calcolo K_c da d_a

$$|e_\infty^{d_s}| = \left| \lim_{t \rightarrow \infty} e_{d_a}(t) \right| = \left| \lim_{t \rightarrow \infty} y_{d_a}(t) \right| = \left| \lim_{s \rightarrow 0} s G_{d_a y}(s) d_a(s) \right|$$

$$|K_c| \geq \left| \frac{D_{a,0}}{\varepsilon_{d_a} G_a G_s G_F} \right| \quad \text{se } \varepsilon_{d_a} \neq 0 \Rightarrow \nu \geq h_a$$

$$|K_c| = \forall \quad \text{se } \varepsilon_{d_a} = 0 \Rightarrow \nu > h_a$$

d) Calcolo K_c da d_p (polinomiale) o ω_c/M_S^{LF} (sinusoidale)

Polinomiale:

$$|e_\infty^{d_p}| = \left| \lim_{t \rightarrow \infty} e_{d_p}(t) \right| = \left| \lim_{t \rightarrow \infty} y_{d_p}(t) \right| = \left| \lim_{s \rightarrow 0} s G_{d_p y}(s) d_p(s) \right|$$

$$|K_c| \geq \left| \frac{D_{p,0}}{\varepsilon_{d_p} G_a G_s G_F K_p} \right| \quad \text{se } \varepsilon_{d_p} \neq 0 \Rightarrow \nu \geq h_p - p$$

$$|K_c| = \forall \quad \text{se } \varepsilon_{d_p} = 0 \Rightarrow \nu > h_p - p$$

Sinusoidale:

$$|e_\infty^{d_p}| = \left| G_{d_p y}(j\omega_p) a_p \sin(\omega t + \varphi_p) \right| \leq |S(j\omega_p)|$$

$$M_S^{LF} = \frac{\rho_p}{a_p}, \quad M_{S,dB}^{LF} = 20 \log_{10} M_S^{LF}$$

$$\omega_L = \omega_p \cdot 10^{-M_{S,dB}^{LF}/40}, \quad \omega_c \geq 2\omega_L$$

e) Calcolo ω_c/M_T^{HF} da d_s (sinusoidale)

$$|e_\infty^{d_s}| = \left| G_{d_s y}(j\omega_s) a_s \sin(\omega t + \varphi_s) \right| \leq |T(j\omega_s)|$$

$$M_T^{HF} = \frac{\rho_s G_s}{a_s}, \quad M_{T,dB}^{HF} = 20 \log_{10} M_T^{HF}$$

$$\omega_H = \omega_s \cdot 10^{M_{T,dB}^{HF}/40}, \quad \omega_c \leq \frac{\omega_H}{2}$$

f) Calcolo ζ , $T_{p,0}$, $S_{p,0}$ da $\hat{s}$

$$\zeta \geq \frac{|\ln(\hat{s})|}{\sqrt{\pi^2 + \ln^2(\hat{s})}}$$

$$T_{p,0} \leq \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

$$S_{p,0} \leq \frac{2\zeta\sqrt{2+4\zeta^2+2\sqrt{1+8\zeta^2}}}{\sqrt{1+8\zeta^2+4\zeta^2-1}}$$

g) Calcolo ω_c da t_r e $\varepsilon_{s,\%}$

$$t_r : \quad \omega_c \geq \left(\frac{\pi - \arccos(\zeta)}{t_r \sqrt{1-\zeta^2}} \right) \sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}$$

$$t_{s,\alpha\%} : \quad \omega_c \geq \left(\frac{|\ln(\alpha\%)|}{\zeta t_{s,\alpha\%}} \right) \sqrt{\sqrt{1+4\zeta^4} - 2\zeta^2}$$

h) Conclusioni

Scelgo ν più grande tra i ν che ho trovato e prendo il K_c più grande tra quelli che ho trovato con quel ν .

2) Progetto del Sistema di Controllo

Passo 1: Studio del segno di K_c e stabilizzabilità

a. Scelgo un segno e calcolo $L_{in}(s)$:

$$L_{in}(s) = \frac{K_c}{s^\nu} \cdot G_F G_a G_p G_s$$

b. Tracciare il diagramma di Nyquist di $L_{in}(s)$ e calcolo

$P_{cl} = N + P_{ol}$, dove:

N : intersezioni semiretta

P_{ol} : poli con $\text{Re}(p_i) > 0$ (di solito nullo)

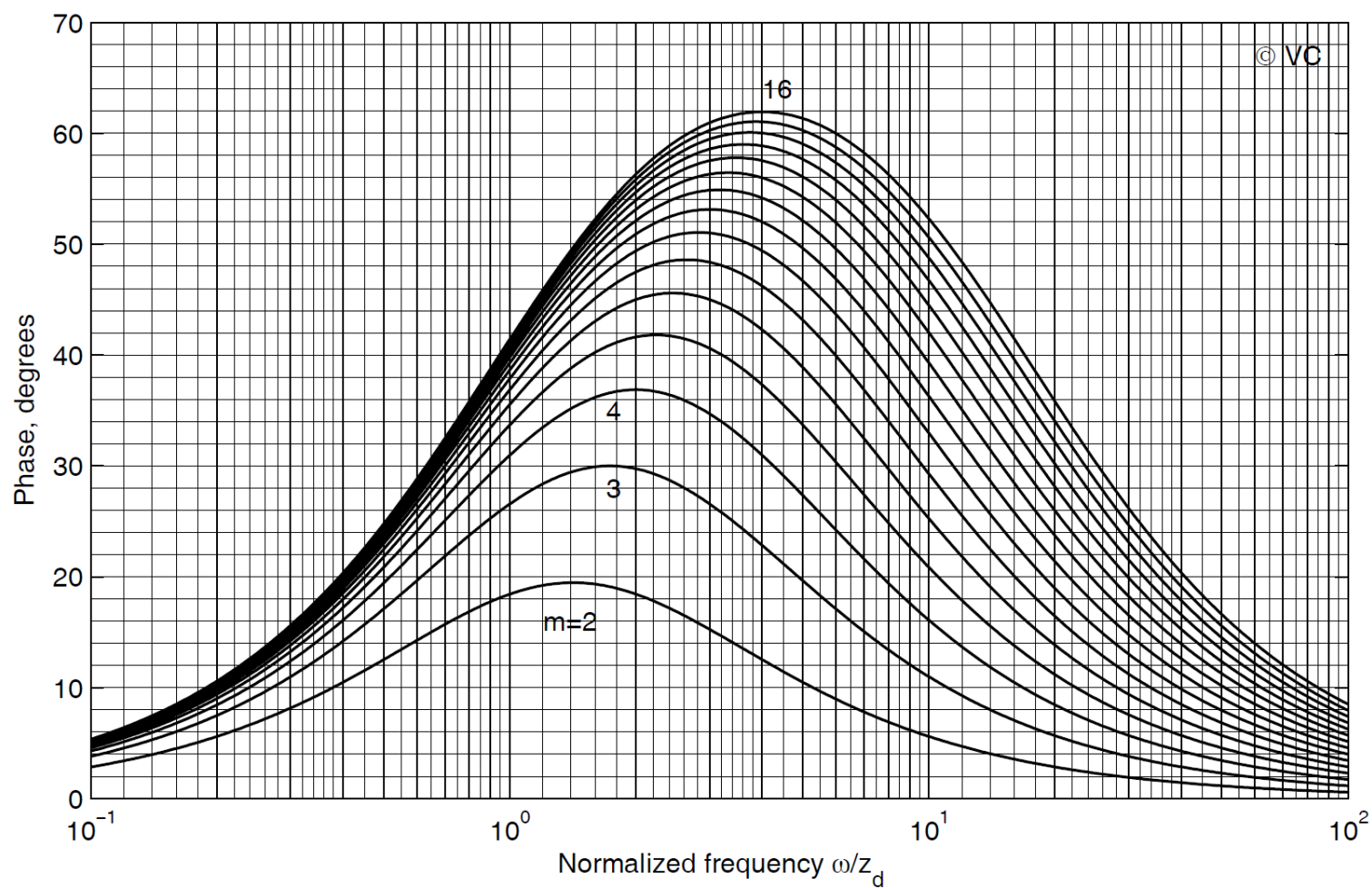
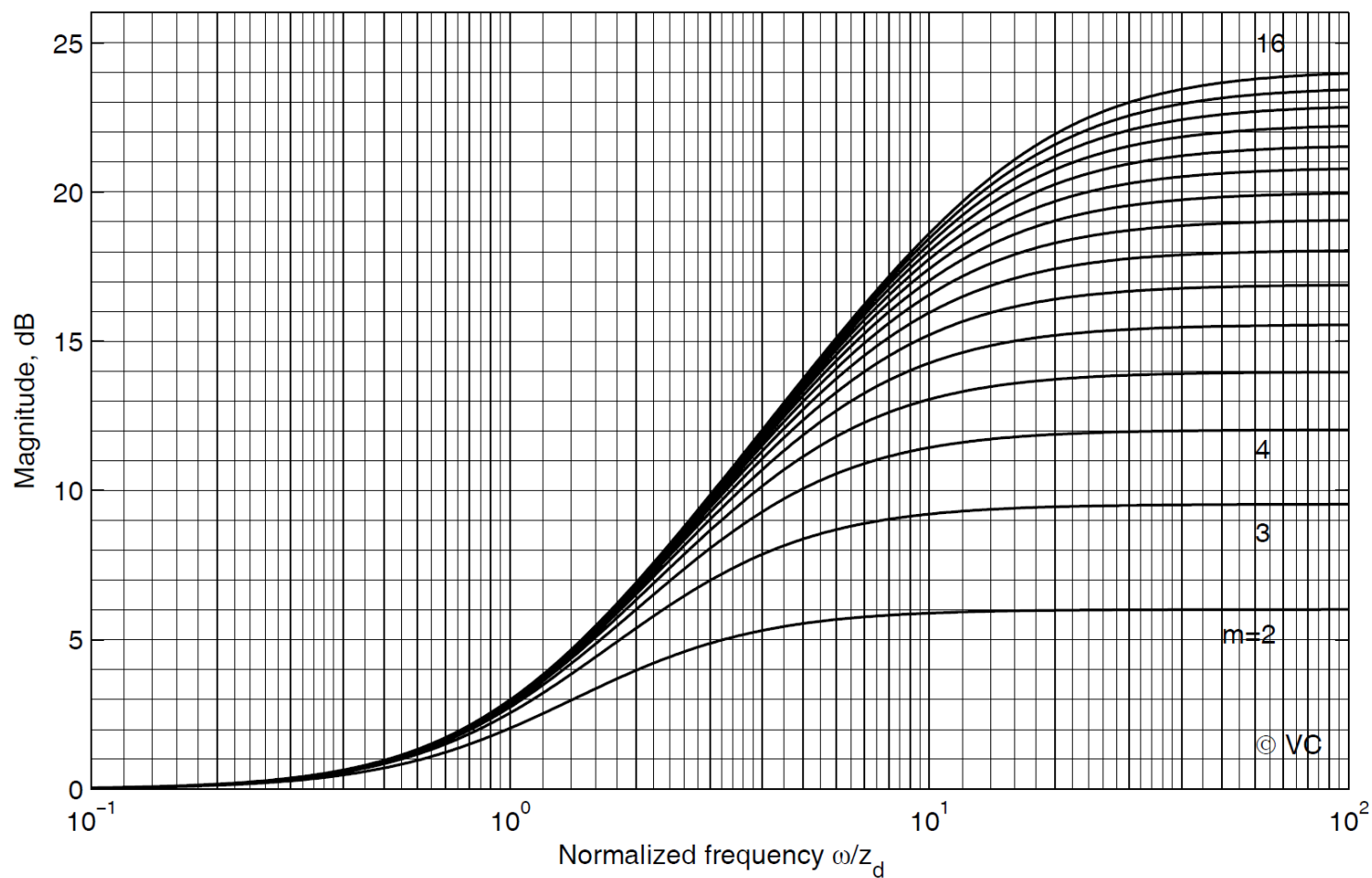
• P_{cl} pari $\rightarrow$ **Stabilizzabile**

• P_{cl} dispari $\rightarrow$ **Cambio segno di K_c**

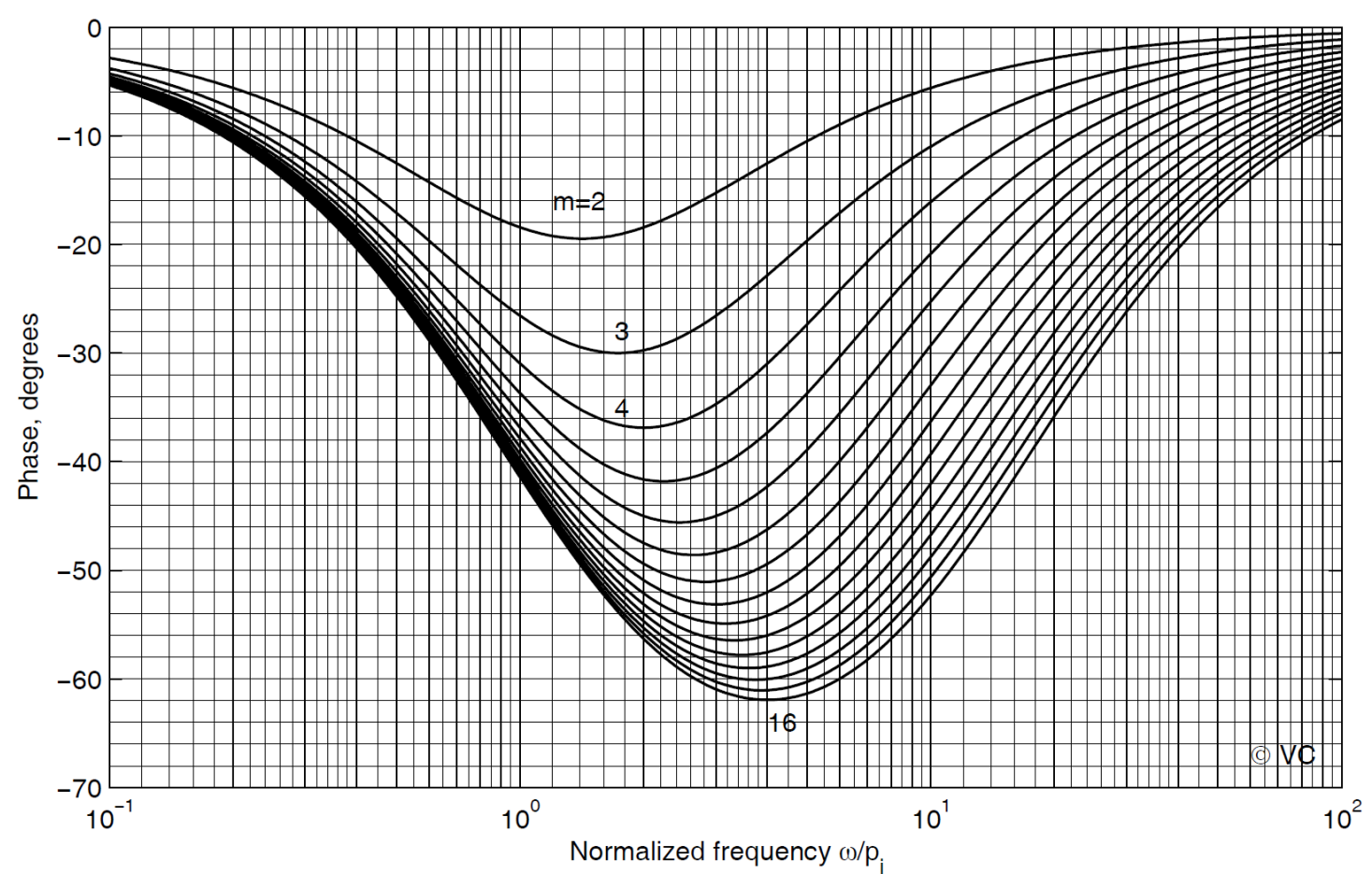
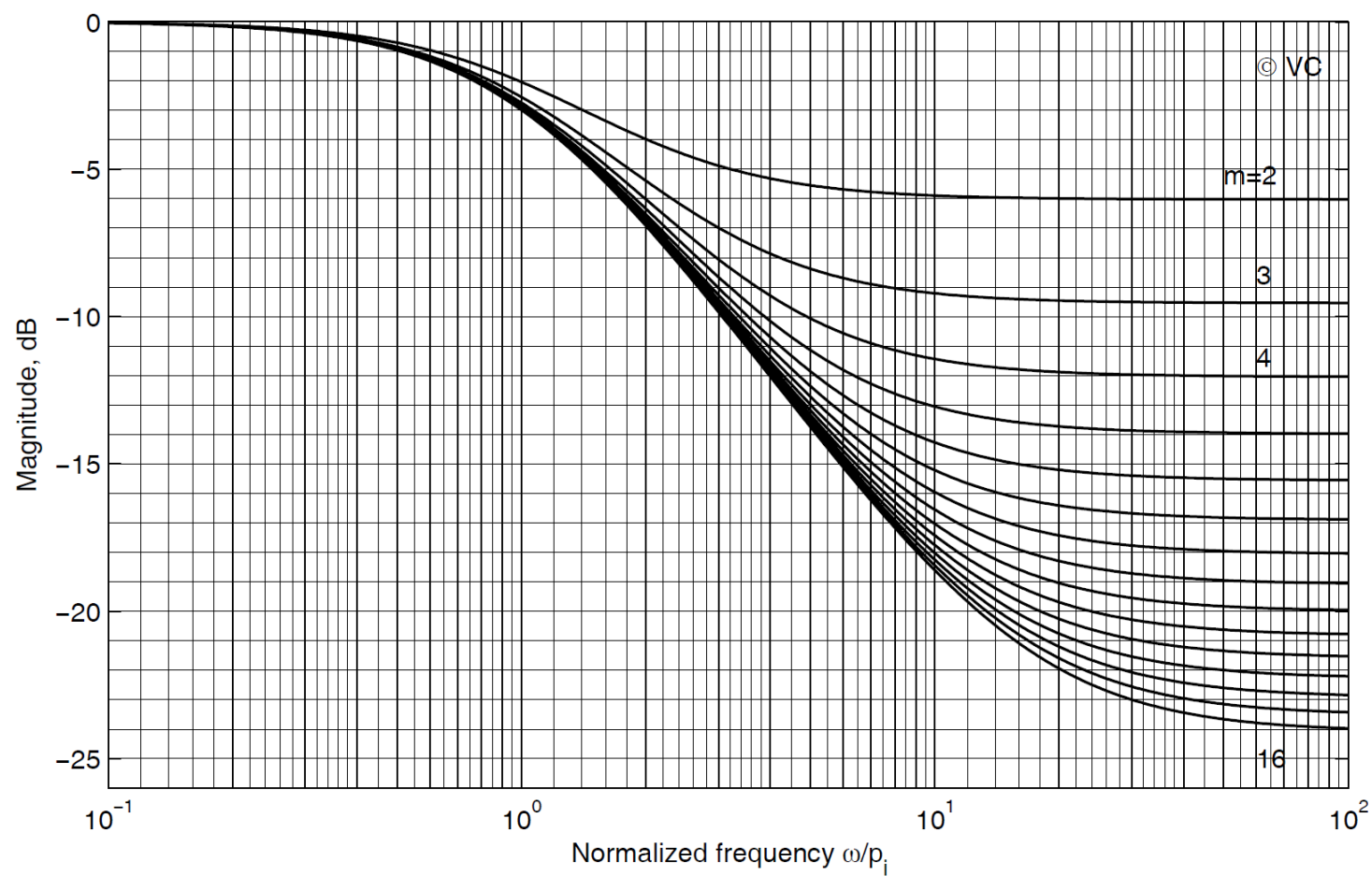
c. Scelgo K_c poco più grande del vincolo (se c'è).

$$\begin{aligned} [N_{L_{in}}, D_{L_{in}}] &= \text{tfdata}(L_{in}, 'v') \\ \text{mynggridst}(T_{p,0}, S_{p,0}) \\ \text{nyquist1}(N_{L_{in}}, D_{L_{in}}) \end{aligned}$$

Phase LEAD network



Phase LAG network



Zero network

